

Computer Algebra (SCI19 3112) — Final Cheat Sheet

Groups/Euler/crypto · Complex/polar/roots · Iso · FTA/fields · Coeff/value · FFT/IFFT · GCD/systems

1 — Groups Recap + Euler φ

Group: closure, identity e (unique, $a \circ e = a$), associativity, inverses (unique). $(\mathbb{Z}, +)$ infinite group; $(\mathbb{Z}_n, +)$ finite, always cyclic (gen $[1]$); $\mathbb{Z}_n^*$ group $\iff n$ prime. $|G| = \# \text{elem}$; $\text{ord}(g) =$ smallest $n > 0$ with $g^n = e$.

Lagrange: $\text{ord}(g) \mid |G|$, $g^{|G|} = e$

Cyclic: $G = \langle g \rangle \iff \text{ord}(g) = |G|$

Worked: $|\mathbb{Z}_8^+| = 8$; $\text{ord}[1] = 8$; $\text{ord}[2] = 4$ (2, 4, 6, 0); $\text{ord}[7] = 8$ (7, 6, 5, 4, 3, 2, 1, 0); gens $[1], [3], [5], [7]$. $\varphi(n) = \# \{1 \leq k \leq n : \gcd(k, n) = 1\}$. **Formulas:** p prime $\varphi(p) = p-1$; $\varphi(p^k) = p^k - p^{k-1}$; coprime $\varphi(mn) = \varphi(m)\varphi(n)$.

Worked: $\varphi(16) = 8$ (1, 3, 5, 7, 9, 11, 13, 15);

$\varphi(17) = 16$.

$\varphi(14) = \varphi(2)\varphi(7) = (2-1)(7-1) = 1 \cdot 6 = 6$

$\varphi(9) = 9 - 3 = 6$ (prime power)

Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$ (= Lagrange on units mod n , $|G| = \varphi(n)$). FLT = prime case ($\varphi(p) = p-1$) = Lagrange on $\mathbb{Z}_p^*$. **Use:** prime $\rightarrow$ FLT (smaller exponent); composite $\rightarrow$ Euler. **Idea:** reduce exponent mod $\varphi(n)$.

Worked $3^{347} \bmod 14$: Given $\gcd(3, 14) = 1$,

$\varphi(14) = 6$.

Step 1 (reduce exponent): $347 = 6 \cdot 57 + 5$ (6:57=342).

Step 2 (evaluate): $3^5 = 243$;

$243 - 17 \cdot 14 = 243 - 238 = 5 \Rightarrow [5]$. **Check:** $\text{pow} = 5$.

2 — Crypto

Caesar: $c \equiv m + s \pmod{26}$, $m \equiv c - s$. **Worked**

BY (B=1, Y=24), $s=20$: $1+20=21=\text{V}$,

$24+20=44-26=18=\text{S} \Rightarrow \text{VS}$.

Power: $c \equiv m^b \pmod{p}$; $m \equiv c^d$, $bd \equiv 1$

(mod $p-1$). **Worked** $p=29$, $b=5$: B: $1^5=1=\text{B}$.

Y **Step 1** ($24 \equiv -5$): $(-5)^2 = 25$; **Step 2:**

$25^2 = 625$; $625 - 21 \cdot 29 = 625 - 609 = 16$.

Y **Step 3:** $16 \cdot (-5) = -80$;

$-80 + 3 \cdot 29 = -80 + 87 = 7 \Rightarrow \text{H}$. d (test d or Euclid):

$5d \equiv 1 \pmod{28} \Rightarrow d=17$ ($5 \cdot 17 = 85 = 3 \cdot 28 + 1$).

Correctness = bijective: $D(E(x)) = x$,

$E(D(y)) = y$. Caesar always bijective (shift). Power

bijective $\iff \gcd(b, p-1) = 1$ (else many-to-one).

Many-to-one: ≥ 2 symbols map together;

surjective: covers whole target.

Check decrypt: VS-20: $21-20=1=\text{B}$,

$18-20=-2+26=24=\text{Y}$. Power: $7^{17} \bmod 29 = 24=\text{Y}$

Check OK.

3 — Complex Numbers

Rect $z = a + bi$ ($\text{Re} = a$, $\text{Im} = b$), plane vector (a, b) .

Polar $r(\cos \theta + i \sin \theta)$, $r = |z| = \sqrt{a^2 + b^2}$. $|z|^2 = z\bar{z}$,

$\bar{z} = a - bi$.

$+$, $-$ = vector + (parallelogram). Mult:

$(a+bi)(c+di) = (ac-bd) + (ad+bc)i$.

Worked 3+4i: $|z| = 5$,

$z^{-1} = (3-4i)/25 = 0.12 - 0.16i$.

Worked (1+6i)/(1-6i):

$(1+6i)^2/37 = (-35+12i)/37$.

$\approx -0.9459 + 0.3243i$ **Check** $(37 \cdot (-0.9459)) = -35$,

$37 \cdot 0.3243 = 12$).

Euler: $e^{i\theta} = \cos \theta + i \sin \theta$; **DeMoivre:**

$(re^{i\theta})^n = r^n (\cos n\theta + i \sin n\theta)$.

n -th roots: $\sqrt[n]{r}(\cos(\theta+2k\pi)/n + i \sin(\theta+2k\pi)/n)$, $k=0..n-1$.

Div: $z_1/z_2 = z_1\bar{z}_2/|z_2|^2$; **Inverse:** $z^{-1} = \bar{z}/|z|^2$.

Polar mult: $r_1 r_2 e^{i(\theta_1 + \theta_2)}$;

$e^{i\pi/3} \cdot e^{i\pi/6} = e^{i\pi/2} = i$.

Worked $8e^{i\pi/3}$: $8(\frac{1}{2} + i\frac{\sqrt{3}}{2}) = 4 + 4\sqrt{3}i \approx 4 + 6.9282i$.

Worked $11e^{3\pi i/2}$: $11(0-i) = -11i$.

Cube roots of $e^{i\pi/2}$: angles

$\pi/6, 5\pi/6, 3\pi/2 \Rightarrow \sqrt{3}/2 + i/2, -\sqrt{3}/2 + i/2, -i$.

Square roots of $e^{i\pi/4}$: angles $\pi/8, 9\pi/8$:

$\cos \frac{\pi}{8} = \sqrt{2+\sqrt{2}}/2 \approx 0.9239$,

$\sin \frac{\pi}{8} = \sqrt{2-\sqrt{2}}/2 \approx 0.3827$.

Rect $\rightarrow$ polar: $2+2i$: $r = \sqrt{8} = 2\sqrt{2}$,

$\theta = \pi/4 \Rightarrow 2\sqrt{2}e^{i\pi/4}$ (**Check**

$2\sqrt{2}(\sqrt{2}/2 + i\sqrt{2}/2) = 2+2i$).

Geometry: $\times e^{i\theta}$ rotates by θ (lengths mult); $\bar{z}$

mirrors across real axis; unit circle $\{|z|=1\}$

subgroup (closed $|zw|=1$, inv $\bar{z}$). **Plot recipe:**

mark Re on x , Im on $y \rightarrow$ arrow; add tip-to-tail;

mult adds angles, mults lengths.

Groups: $(\mathbb{C}, +)$ $e=0$ inv $-z$; $(\mathbb{C}^*, \cdot)$ $e=1$ inv $\bar{z}/|z|^2$;

$C_4^* = \{\pm 1, \pm i\}$ gen i ; C_8^* gen $(1+i)/\sqrt{2}$; $\{1, -1\}$

group (gen -1); $\{i, -i\}$ **not** ($i \cdot i = -1$ missing).

Mult prop $|z_1 z_2| = |z_1| |z_2|$: $z_1 = (1+i)/\sqrt{2}$,

$z_2 = (1-i)/\sqrt{2}$: $1 \cdot 1 = |1| = 1$ **Check OK**.

4 — Roots of Unity

Powers $\omega^0.. \omega^{n-1}$ = all n roots cyclically (regular

n -gon on unit circle, angle $2\pi/n$ apart); primitive

= generator; n distinct roots of $z^n - 1 = 0$.

Primitive: ω_k^n generates all $\iff \gcd(k, n) = 1$;

e.g. $n=8$: $k=1, 3, 5, 7$. **Why:**

$\text{ord}(\omega_k^n) = n / \gcd(k, n)$ — shared factors shortcut

the cycle (e.g. $\omega_8^2 = \omega_4$, order 4). **Gen test:** $g^d \neq 1$

for all proper divisors $d \mid n$ (Z13:

$2^1, 2^2, 2^3, 2^4, 2^6 = 2, 4, 8, 3, 12 \neq 1$).

Cyclic: $C_n^* = \langle \omega_n \rangle$; reduce first: $\omega_8^2 = \omega_4 = i$.

Root: $\omega_n = e^{2\pi i/n}$.

Relations (n even; last two $\forall n$): $\omega_n^n = 1$, $\omega_n^{n/2} = -1$, $\omega_n^2 = \omega_{n/2}$, $\omega_n^{-1} = \bar{\omega}_n$, $\omega_n^{k+n/2} = -\omega_n^k$, $\omega_n^{2k} = \omega_{n/2}^k$, $\omega_n^{-k} = \bar{\omega}_n^k$.

Worked $\omega_8 = (1+i)/\sqrt{2}$: $\omega_8^4 = -1$, $\omega_8^8 = 1$,

$\omega_8^{-5} = \omega_8^{-5+8} = \omega_8^3 = (-1+i)/\sqrt{2}$;

$\omega_{16}^8 = e^{i\pi/2} = i = \omega_4$. $\sqrt{i} = (1+i)/\sqrt{2}$ (**Check**

$((1+i)/\sqrt{2})^2 = i$); $\sqrt{-i} = (-1+i)/\sqrt{2}$.

Eval trick: reduce exponents mod n , use $\omega_2 = -1$.

Worked $p = x^4 - 3x^3 + x^2 - 2x - 1$ at $\omega_4 = i$: powers

$\omega_4^2 = -1$, $\omega_4^3 = -i$, $\omega_4^4 = 1$.

Then

$1 - 3(-i) + (-1) - 2i - 1 = 1 + 3i - 1 - 2i - 1 = -1 + i$.

Lecture: $3\omega_4^3 - 2\omega_4^2 + 2 + \omega_4^8 = 3(-i) + 2 + 2 + 1 = 5 - 3i$.

5 — Iso $C_n^* \cong \mathbb{Z}_n^+$

Map $\omega_n^k \leftrightarrow [k]_n$; $\times \leftrightarrow +$:

$\omega_n^k \omega_n^l = \omega_n^{k+l} \leftrightarrow [k] + [l] = [k+l]$. Gen $\leftrightarrow$ gen. **Isos:**

bijection preserving op (C_n^* table $\equiv \mathbb{Z}_n^+$ table).

Subgroup: subset + closed under op.

Worked $C_4 \leftrightarrow \mathbb{Z}_4$: $[0] \leftrightarrow 1, [1] \leftrightarrow i, [2] \leftrightarrow -1, [3] \leftrightarrow -i$;

$[1] + [2] = [3] \leftrightarrow i \cdot (-1) = -i$;

$[3] + [2] = [5] = [1] \leftrightarrow (-i) \cdot (-1) = i$. Tables same shape

$\Rightarrow$ iso.

No-iso test: $|G|$ must match + generator orders;

e.g. $|C_8^*| = 2$ vs $|\mathbb{Z}_4^+| = 4 \Rightarrow$ no iso. **Fill C_5^* via iso:**

$\omega_3^3 \cdot \omega_5^4 = \omega_5^2 = \omega_5^2 \leftrightarrow [3] + [4] = [7] = [2]$. Subgroups:

$C_{2m}^* \subset C_{2m+1}^*$ ($z^{2m} = 1 \Rightarrow (z^{2^m})^2 = 1$).

6 — Poly: FTA, Fields, Coeff/Value

Field: $+$, $-$, $\times$, $/$ (no $/0$); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ yes; $\mathbb{Z}$ no (a/b

can leave $\mathbb{Z}$); $\mathbb{N}$ no ($a-b$ can leave $\mathbb{N}$); $\mathbb{Z}_p$ (p prime)

yes (every $\neq 0$ invertible). Polys over field K

(usually $\mathbb{R}/\mathbb{C}$); some fields give no roots. Alg

closed: every $\deg \geq 1$ has a root; $\mathbb{C}$ is (FTA); $\mathbb{R}$ is

not ($x^2 + 1 = 0$ no real root).

FTA: $\deg n \Rightarrow$ **exactly n roots in $\mathbb{C}$** counting

mult: $f = a_n \prod (x - x_i)$.

Worked $x^4 + 16 = 0$: no real roots; 4 in $\mathbb{C}$; two:

$\sqrt{2} + i\sqrt{2}, -\sqrt{2} + i\sqrt{2}$ (**Check** $(\sqrt{2} + i\sqrt{2})^2 = 4i$,

$^2 = -16$). All four: $\pm \sqrt{2} \pm i\sqrt{2}$ (signs independent;

$|x|=2$, angles $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$).

4th roots of -1 : $-1 = e^{i\pi}$: angles

$(\pi + 2k\pi)/4 = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4 \Rightarrow (\pm 1 \pm i)/\sqrt{2}$

= odd ω_8 powers (all primitive).

Proof (no real roots): for real x , $x^4 \geq 0$ (even

power), so $x^4 + 16 \geq 16 > 0$ — never 0. I.e. $x^4 = -16$

impossible over $\mathbb{R}$ ($\text{LHS} \geq 0$). Hence all 4 FTA

roots are nonreal. Factor thm:

$f(a) = 0 \iff (x-a) \mid f$; $(x-1)^2 = x^2 - 2x + 1$

(multiple root); $x^2 + 1 = 0$: roots $\pm i$ (none real).

Note: 5 prime in $\mathbb{Z}$ but $5 = (2-i)(2+i)$ in $\mathbb{Z}[i]$.

Coeff ($a_0..a_n$); **value:** $\deg \leq n$ unique from $n+1$

pts (Vandermonde $\det \neq 0$). **Product deg $n+m$**

needs $n+m+1$ pts; pad zeros

$(4+5x \rightarrow 4+5x+0x^2)$. Pointwise

$h(x_i) = f(x_i)g(x_i)$, $O(m)$; schoolbook $O(n^2)$.

Worked $f=1-x, g=2+x$ at $-2, 0, 1$:

$f: 1 - (-2) = 3, 1, 0$; $g: 2 + (-2) = 0, 2, 3$;

$h: 0, 2, 0 \Rightarrow h = 2 - x - x^2$ (**Check** $2 + 2 - 4 = 0, 2,$

$2 - 1 - 1 = 0$).

Sketch recipe: mark roots $\rightarrow$ lead-coeff sign for

ends $\rightarrow y$ -intercept $\rightarrow$ join; graphical mult: sample

≥ 6 x 's, multiply y 's, plot.

Point-count recipe: single $\deg-n \rightarrow n+1$ pts;

product $\deg n+m \rightarrow n+m+1$; FFT $\rightarrow$ next pow2

$\geq$ that.

7 — FFT/IFFT + Costs

Even/odd: $p(x) = p_e(x^2) + x p_o(x^2)$. **Worked**

$1 - 2x + 3x^2 - 4x^3 + 5x^4 - 6x^5 + 7x^6 - 8x^7$:

$p_e(y) = 1 + 3y + 5y^2 + 7y^3$, $p_o(y) = -2 - 4y - 6y^2 - 8y^3$;

depth of recursion tree = 3 ($8=2^3$).

Butterfly ($k=0..n/2-1$):

$p(\omega_n^k) = p_e(\omega_n^{2k}) + \omega_n^k p_o(\omega_n^{2k})$;

$p(\omega_n^{k+n/2}) = p_e(\omega_n^{2k}) - \omega_n^k p_o(\omega_n^{2k})$. Needs $n=2^t$.

Butterfly $n=4$ ($\omega = i$):

$k=0$: $p(1) = p_e(1) + p_o(1)$

$p(-1) = p_e(1) - p_o(1)$

$k=1$: $p(i) = p_e(-1) + i p_o(-1)$

$p(-i) = p_e(-1) - i p_o(-1)$

$N_{\text{FFT}} \geq 2n-1$, smallest pow2 (4+4 coeffs $\deg 3$:

product $\deg 6$, 7 coeffs $\Rightarrow N=8$). Pipeline:

coeff $\rightarrow$ FFT $\rightarrow$ vals $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ coeff.

Zero-pad; extra points OK if same poly. **Idea:** one

(p_e, p_o) pair gives two values (halves work).

IFFT: $a_j = \frac{1}{n} \sum_{k=0}^{n-1} y_k (\omega_n^{-j})^k$, $j=0..n-1$.

Costs: FFT/IFFT $O(n \log n)$ (depth

$\log n \times O(n)$ /level); pointwise $O(n)$; total

$O(n \log n)$.

Worked FFT $q = 1 - x^2 + 4x^3</$

[2, 15, 7] → [2, 5, 8]=852 (raw 2+150+700=852; notes typo 825). 5678 → 8+7x+6x²+5x³; q(10)=5678, q(2)=86 (digits < 10, no carry needed). d decimals → ⌈d/3⌉ base-1000 digits: 120!: 199 → 67 (66·3=198<199). **Idea**: every digit must land in [0, B−1] (carry the rest). **GCD monic** (lead 1; unique up to ×const): 4x²+12x−7 → x²+3x−7/4; gcd(x²−1, x−1)=x−1 ((x−1)(x+1)). Euclid via synthetic division. **Worked** f=x⁴−8x³+17x²+2x−24, g=x³−6x²+11x−6: **Step 1** (divide, two terms): x⁴/x³=x, then −2x³/x³=−2: f−(x−2)g=−6x²+30x−36=−6h, h=x²−5x+6. **Step 2**: g rem h is 0 ⇒ gcd(f, g)=h=x²−5x+6. **Synthetic** (x³−6x²+11x−6)/(x−1): bring 1; ×1: 1 → −5; ×1: −5 → 6; ×1: 6 → 0. Quot x²−5x+6, rem 0 (so x=1 root). **Overdetermined**: h=gcd; deg h=0 ⇒ inconsistent (x−1=0, x+1=0); deg ≥1 ⇒ common roots = roots of h. Mini: x²−1=0 (±1) + x−1=0 (1) ⇒ common x=1. Lecture consistent pair: x−1=0, 2x−2=0 (same root 1, h=x−1). **Worked**: x²−11x+30=0: 25−55+30=0, 36−66+30=0 ⇒ 5, 6. x³−10x²+27x−18=0: test 6: 216−360+162−18=0; roots 1, 3, 6 ⇒ common **x=6**. **Resultant**: det condition for common root (eliminates a variable). 2×2: Res(x−1, x+1)=(1)(1)−(−1)(1)=2≠0 ⇒ no common root.

9 ————— **Worked Solutions (Step-by-Step)**
Extra **Q&A (Quiz/HW Style)**

FFT size Qs: p with n coeffs needs n points forth+back; FFT needs pow2 so pad. n−1=16 ⇒ (not pow2) ⇒ **cannot use directly**, pad to 32. (p(x))²: deg 2(n−1), needs 2n−1 pts → next pow2. **IFFT** n=2, y=(1, 1+i): x₀=1=ω₂⁰, x₁=−1=ω₂¹; a₀=(y₀+y₁)/2=(2+i)/2=1+0.5i; a₁=(y₀−y₁)/2=−0.5i. **Check**: p(1)=1, p(−1)=1+i. Gen a₂ formula (n=4): a₂=(y₀−y₁+y₂−y₃)/4 (ω^{−2k}=1, −1, 1, −1). **Idea**: IFFT = FFT with ω^{−1}, then /n (**trap**: forgetting /n or the −1). **C₈^{*} products** (z₁=1, z₂=−1, z₅=(1+i)/√2, z₆=(−1+i)/√2, z₇=(1−i)/√2, z₈=(−1−i)/√2): z₂z₅=(−1−i)/√2=z₈; z₅z₇=1=z₁; z₆z₈=1=z₁. Inverse of (−1−i)/√2: |z|²=1, inv = $\bar{z}$ =(−1+i)/√2=z₆. **Group tests**: 2a+7bi (a, b∈ℤ) additive: closed, e=0, inv −z ⇒ **group**. 3a+3bi, ab≠0 multiplicative: 1=3a+3bi ⇒ a=1/3∉ℤ ⇒ **not** (no identity). **FFT** n=32: depth 5 (2⁵=32); butterfly pairs (ω⁰, ω¹⁶), (ω¹, ω¹⁷); even/odd parts deg 15, split again → 4 polys (p_{ee}, p_{eo}, p_{oe}, p_{oo}). C_n⁺ = cyclic additive order n (≅ℤ_n⁺), gens [k] with gcd(k, n)=1; chain C_{2m}^{*}⊂C_{2m+1}^{*}. Formulas: p(ω⁰)=p_e(1)+p_o(1), p(ω¹⁶)=p_e(1)−p_o(1); p(ω¹)=p_e(ω²)+ω p_o(ω²), p(ω¹⁷)=p_e(ω²)−ω p_o(ω²). **Inconsistent check**: gcd(x−1, x+1)=1 (deg 0) ⇒ **inconsistent** ((x+1)−(x−1)=2≠0).

————— **Mod + Groups/FLT Drills**

Reduce first: [125]₁₇+ [241]₁₇= [366]₁₇; 366−21·17=366−357=9. [391]₂₁·[457]₂₁: 391−18·21=13, 457−21·21=16; 13·16=208−9·21=19. [94]₇: 94−13·7=94−91=3. ([157]₁₃+ [249]₁₃)·[83]₁₃: 157−12·13=1,

249−18·13=249−234=15; 15−13=2; 1+2=3; 83−6·13=5; 3·5=15−13=2. **HW08 reduce-first**: [123]₁₇+ [245]₁₇=11 (368−357). [391]₂₃·[457]₂₃=0 (391=17·23). ([156]₁₁+ [247]₁₁)·[83]₁₁: 2+5=7, ×6=42−33=9. [312]₁₅·([184]₁₅+ [257]₁₅): 12·(4+2)=72−60=12. **Find x**: 143 ≡ x (mod 13) : 143=11·13 ⇒ 0 12345 mod 11 : 11·1122=12342 ⇒ 3 54321 mod 7 : 7·7760=54320 ⇒ 1 147 mod 13 : 13·11=143 ⇒ 4 **Find n**: 24 ≡ 0 (mod n) ⇒ n|24: 1, 2, 3, 4, 6, 8, 12, 24. 15 ≡ 1 ⇒ n|14: 1, 2, 7, 14. 124 ≡ 4 ⇒ n|120, n>4: 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. 18 ≡ 0, 0<n≤18: 1, 2, 3, 6, 9, 18. **Classes**: mod 7: 7 classes [0]..[6]. [2]₅= { ..., −8, −3, 2, 7, 12, ... }; [4]₇= { ..., −3, 4, 11, ... }; [−3]₈= [5]₈= { ..., −3, 5, 13, ... }. {1, −1}⊂C₄^{*} subgroup (closed, e=1, self-inv). **Notation ↔**: [8]₁₁+ [9]₁₁= [6]₁₁ ⇔ 8+9 ≡ 6 (mod 11) (17−11=6); [4]₇·[5]₇= [6]₇ ⇔ 4·5 ≡ 6 (20−14=6); 17+9 ≡ 2 (mod 8) ⇔ [17]₈+ [9]₈= [2]₈ (26−24=2); 6·8 ≡ 3 (mod 5) ⇔ [6]₅·[8]₅= [3]₅ (48−45=3). **Groups**: (Q, +) group; (Q, ·) **not** (0 no inverse). (ℤ₄, +) group (e=[0], inv [−a]). ℤ₄^{*}= {1, 2, 3} **not** (2·2=0). Mult groups — which n admit one? n=3, 5, 13, 17 yes (|G|=2, 4, 12, 16); 10, 15 no. Find gens: ℤ₅^{*} gens {2, 3} (ord=4). Possible orders (divisors of |G|): 2→{1, 2}; 4→{1, 2, 4}; 12→{1, 2, 3, 4, 6, 12}; 16→{1, 2, 4, 8, 16}. |ℤ₂₄^{*}|=φ(24)=8 (odd, not mult of 3: 1, 5, 7, 11, 13, 17, 19, 23); x odd (e.g. 3, 5, 7) never gives [a]^x= [1] for a=2..23 (needs 2|x; [5]²= [1] so every even x works). **Gen of ℤ₁₃^{*}**: |G|=12, orders ∈ {1, 2, 3, 4, 6, 12}; 2²=4, 2³=8, 2⁴=3, 2⁶=12≠1 ⇒ ord(2)=12 ⇒ generator. **FLT big powers**: 19⁶ mod 7 OK (7∤19); 21⁶ **no** (gcd=7); 11⁸ mod 9 **no** (9 composite). 19⁶⁸: 19 ≡ 5, 68=6·11+2 (66+2), 5²=25 ≡ 4. 271²⁴ mod 11: 271−24·11=271−264=7, 24=10·2+4, 7⁴=2401−218·11=2401−2398=3. 290²⁹³ mod 23: 290−12·23=290−276=14, 293=22·13+7 (286+7), 14⁷ ≡ 19. **Simplify without FLT**: [16⁴]₇= [2]; [10¹⁰⁰]₉= [1]; [23²]₉= [7] (23 ≡ 5, 25−18); [42]₁₂= [6]. **More Euler**: 5¹⁰⁰ mod 13: φ=12; 100=12·8+4; 5⁴=625−48·13=625−624=1. 2¹⁰⁰⁰ mod 7: 1000=6·166+4; 2⁴=16 ≡ 2. **Compute φ**: φ(8)=8−4=4; φ(24)=φ(8)φ(3)=4·2=8; φ(100)=φ(4)φ(25)=2·20=40. **Trap**: φ(4)φ(2)=2·1=2≠4=φ(8) (gcd(4, 2)≠1). **More cipher**: encrypt HI (s=20): 7+20=27−26=1=B, 8+20=28−26=2=C ⇒ **BC**. Power m=2 (p=29, b=5): 2⁵=32−29=3. **Trap**: 2⁴ mod 8=0≠1 though φ(8)=4 (gcd≠1). **Mod x³−1 (Quiz4 Q6)**: x³ ≡ 1: [2x⁵−9x³+5x+9]=2x²−9+5x+9=[2x²+5x]. Square after reducing: [(2x²+5x)²]=[4x⁴+20x³+25x²]; x⁴ ≡ x, x³ ≡ 1 ⇒ [25x²+4x+20].

————— **Complex + FFT Drills**

Sums (plot tip-to-tail, check): (−2+4i)+(−3−7i)=−5−3i (−2−i)+(2+i)=0 (1−2i)+(1+2i)=2 |z|²=z $\bar{z}$: (−2−3i): 4+9=13; 4i: 16; $\bar{z}$ =4i ⇒ z=−4i, |z|²=16. **Div**: (2+6i)/2=1+3i.

(3+4i)/(1−6i)=(3+4i)(1+6i)/37=(−21+22i)/37. ≈−0.5676+0.5946i **Check** (37·(−0.5676)=−21, 37·0.5946=22). **vs dot**: z₁z₂=(ac−bd)+(ad+bc)i vs u·v=ac+bd; |z₁z₂|²=(a²+b²)(c²+d²) vs (u·v)²=(ac+bd)²; coincide ⇔ ad=bc. E.g. z₁=1+2i, z₂=3+4i: |z₁z₂|²=(1+4)(9+16)=125 vs (3+8)²=121 (1·4≠2·3). **Even/odd twice of p_e=1+3y+5y²+7y³**, p_o=−2−4y−6y²−8y³: p_{ee}(z)=1+5z, p_{eo}(z)=3+7z p_{oe}(z)=−2−6z, p_{oo}(z)=−4−8z **Big-O tree** (n leaves, balanced binary): depth log₂ n; bottom-only work ⇒ O(n); O(n) per level ⇒ O(n log n). **C₅^{*} row via iso**: ω₅⁴·ω₅⁴=ω₅⁸=ω₅³ ↔ [4]+[4]=[8]=[3]. Gen ω₅ ↔ [1]₅ powers fill table.

————— **Tables + FFT Drills**

Mod-8 × table (nonzero rows): Rule: row a has 1 ⇔ a invertible; row a hits 0 ⇒ zero divisor. Units = odd {1, 3, 5, 7} (gcd=1); 1^{−1}=1, 3^{−1}=3 (9−8=1), 5^{−1}=5 (25−24=1), 7^{−1}=7 (49−48=1). Even rows lack 1. **Zero divisors**: 2·4=8 ≡ 0, 6·4=24 ≡ 0 (24−24). **FFT** f=2+3x+5x³ (N=4, ω₄=i; p_e(y)=2, p_o(y)=3+5y): f(1)=2+3+5=10; f(i)=2+3i+5(−i)=2−2i; f(−1)=2−3−5=−6; f(−i)=2−3i+5i=2+2i. **Butterfly table (f)**: k=0: p_e(1)=2, p_o(1)=8 ⇒ f(1)=2+8=10, f(−1)=2−8=−6. k=1: p_e(−1)=2, p_o(−1)=−2 ⇒ f(i)=2+i(−2)=2−2i, f(−i)=2−i(−2)=2+2i. **DeMoivre power**: ((1+i)/√2)⁸=(e^{iπ/4})⁸=e^{2πi}=1 (angle ×8). **Decrypt proof**: m ≡ c^d=(m^b)^d=m^{bd}=m^{1+k(p−1)}=m·(m^{p−1})^k ≡ m (FLT, p prime, p∤m).

————— **Tables + Step-by-Step Recoveries**

Mod-8 × table (read inverses/zero divisors; odd rows permute, even rows hit 0):

×	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

C₄^{*} vs ℤ₄⁺ (same shape ⇒ iso ω₄^k ↔ [k]₄):

*	1	i	−1	−i	+	0	1	2	3
1	1	i	−1	−i	0	0	1	2	3
i	i	−1	−i	1	1	1	2	3	0
−1	−1	−i	1	i	2	2	3	0	1
−i	−i	1	i	−1	3	3	0	1	2

C₅^{*} table via iso (ω=ω₅; ωⁱ·ω^j=ω^{(i+j) mod 5}):

*	1	ω	ω ²	ω ³	ω ⁴
1	1	ω	ω ²	ω ³	ω ⁴
ω	ω	ω ²	ω ³	ω ⁴	1
ω ²	ω ²	ω ³	ω ⁴	1	ω
ω ³	ω ³	ω ⁴	1	ω	ω ²
ω ⁴	ω ⁴	1	ω	ω ²	ω ³

Recover h=a+bx+cx² from values (0, 2, 0) at (−2, 0, 1): a−2b+4c=0, a=2, a+b+c=0. **Step 1**: a=2. **Step 2**: b+c=−2, −2b+4c=−2 ⇒ b−2c=1. **Step 3**: subtract: 3c=−3 ⇒ c=−1, b=−1. **Box**: h=2−x−x². **Check**: 2+2−4=0, 2, 2−1−1=0. **Read graph → number**: point (x, y) → x+yi; length r, angle θ → r(cos θ+i sin θ).